

Digital Product Passport — Project Overview

- Full-stack platform to create, publish, and share product passports via QR.
- Stack: FastAPI + PostgreSQL + Alembic · Next.js + TypeScript + MUI + RTK Query.
- Local run: Postgres in Docker; API and web on the host (see root README).

How to run (reviewers)

- 1. Copy .env.example and backend/frontend .env.local.example files
- 2. docker compose up db
- 3. backend: uv sync && alembic upgrade head && seed_* && uvicorn
- 4. frontend: npm i && npm run dev
- 5. Open <http://localhost:3000> and <http://localhost:8000/docs>
- Logins: admin@example.com / admin1234 · editor@example.com / editor1234

Core features

- Auth: JWT access + httpOnly refresh cookie; roles admin | editor
- Products with materials, sustainability, certifications, docs, images
- Publish → stable public UUID + QR; public passport page; scan tracking
- Dashboard and analytics; admin user CRUD

Bonuses implemented

- Soft delete + restore/Undo; search and pagination on product list
- Audit log table + admin /audit UI
- Drag-and-drop uploads; passport versioning (republish, same QR URL)
- Passport PDF export (BackgroundTasks cache)
- Redis cache for dashboard/analytics (optional REDIS_URL)
- MinIO-ready Storage backend (STORAGE_BACKEND=minio)
- Purge job: hard-delete soft-deleted products after retention days

Soft delete & purge

- DELETE sets products.deleted_at (row kept; hidden from lists/passport).
- Restore clears deleted_at (UI Undo snackbar).
- CLI purge removes old soft-deleted rows + stored files:
- APP_ENV=local uv run python -m scripts.purge_deleted_products --dry-run
- Retention: SOFT_DELETE_RETENTION_DAYS (default 30).

Demo walkthrough

- 1. Login as admin → Dashboard / Analytics
- 2. Products → DEMO-001 → Publish panel (versions, QR, republish)
- 3. Soft-delete a product → Undo; open Audit
- 4. Public passport → Download PDF; visit with ?src=qr
- 5. Login as editor → Users and Audit hidden

Docs & tests

- Architecture: docs/ARCHITECTURE.md · Database: backend/docs/database.md
- Overview PDF: docs/DPP_Project_Overview.pdf (this file)
- Backend tests: cd backend && APP_ENV=local uv run pytest

- Frontend: npm test · Playwright: npm run test:e2e (API+seeds up)